

Swapnil Sanjay Joijode

Data Engineer

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Professional Summary

Data Engineer with 3+ years of experience designing scalable Python-based data pipelines, modular transformation frameworks, and maintainable cloud data systems across Snowflake, Spark, Airflow and AWS environments. Skilled in data architecture, workflow orchestration, automated testing, CI/CD and refactoring production pipelines into reliable, well-structured components supporting high-volume operational analytics. Proven track record improving pipeline reliability, reducing reconciliation defects, and delivering production-ready data platforms through cross-functional collaboration with analytics, engineering and business teams. Brings strong software engineering discipline focused on clean code, maintainability, observability and scalable system design.

Work Experience

Data and Analytics Engineer | Kalven Group LLC | United States

December 2025 to Present

- Engineered modular Airflow-orchestrated Spark data pipelines and scalable ELT workflows across Snowflake and Databricks processing 2–3M logistics records monthly, reducing reconciliation defects from approximately 70 to fewer than 15 per cycle while improving production reliability and downstream data trust across 40+ stakeholders.
- Architected reusable dbt transformation frameworks, dimensional models and governed data marts on Snowflake and Databricks SQL, improving maintainability, standardising 50+ KPI definitions and reducing model delivery effort by 2 - 4 hours per request.
- Refactored legacy batch ingestion and transformation workflows into maintainable Airflow DAGs, Spark jobs and AWS Lambda-based pipelines, reducing end-to-end runtime from 90 to 55 minutes while improving scalability, schema evolution support and operational stability for near-real-time analytics.
- Implemented automated unit-style validation checks, schema testing, source freshness monitoring and regression-focused data quality controls using Python, SQL and dbt, reducing manual QA effort from 6 - 7 hours to 2 - 3 hours per cycle.
- Designed scalable semantic and reporting layers on Snowflake and Databricks, consolidating 15+ executive reporting workflows into governed, production-ready data products consumed by 40+ business stakeholders.
- Analyzed parcel-level operational data across 200+ hub and route combinations using SQL and Python, identifying delay clusters that enabled routing changes reducing late deliveries on the five highest-risk routes by 18%.
- Developed near-real-time ingestion pipelines and event-driven operational analytics workflows on AWS, improving data availability for logistics decision-making with sub-hour data freshness targets.
- Collaborated on architecture decisions, technical planning and cross-functional delivery across Snowflake, Spark and AWS environments, contributing to maintainable integration patterns and scalable data platform design.

Data Analyst and Power BI Developer | Vision Group Retail Inc. | United States

August 2024 to November 2025

- Redesigned Snowflake dimensional models, aggregation tables, and semantic-layer inputs, reducing model size from 10 GB to 1 GB and refresh time from 10 to 2 minutes across 500K+ records, demonstrating Snowflake modeling depth critical to data platform migrations and consolidations.
- Engineered Snowflake-backed dbt transformation workflows and curated dimensional data models for merchandising and supply chain operations, maintaining data quality standards and governed metric definitions across 500K+ records per refresh cycle.
- Enforced data quality controls using automated SQL and Python validation across 500K+ records per refresh, reducing source discrepancies from thousands to single-digit mismatches against Azure SQL baselines.
- Deployed Python ETL services and dbt transformation workflows through GitLab CI/CD pipelines with automated testing, rollback procedures and deployment monitoring, improving release reliability and eliminating 12+ manual reporting processes per quarter.
- Delivered 50+ governed business metrics through Snowflake-integrated semantic models, positioning the analytics layer as a governed consumption tier on top of curated warehouse datasets.
- Implemented maintainable CI/CD release workflows with auditability controls, deployment validation and rollback support, improving production stability and reducing release variability across data engineering pipelines.

Data Analyst | TekSolve IT Solutions Inc. | United States

August 2023 to December 2023

- Designed automated SQL and Python data quality validation frameworks across 50K+ monthly HR and payroll records, reducing report error rate from 0.58% to 0.18% and achieving SLA compliance across three consecutive delivery cycles.
- Automated SQL and Python validation checks detecting discrepancies 3 to 5 days earlier than manual review, improving on-time monthly reporting pack delivery while supporting reliable upstream ETL workflows.
- Built star-schema data models for HR and finance, cutting dashboard rebuild requests by 40% through reusable dimensional models and self-service documentation.

Senior Business Analyst | Quantum Phinance Consulting Pvt. Ltd. | India

January 2020 to December 2021

- Delivered automated reporting logic and KPI dashboards for approximately 100 financial assets, improving decision-making efficiency by 30% for risk and finance stakeholders.
- Collaborated with business and operational stakeholders to gather requirements, support technical planning, and improve reporting process standardization across finance workflows.

Education

M.S., Data Analytics (Information Management) | Clark University | United States January 2022 to December 2023

- Grade: 3.4/4.0
- Relevant Modules: Database Management, Machine Learning, Statistics, Data Visualisation

B.E., Electronics and Telecommunication Engineering | Mumbai University | India August 2015 to July 2019

- Grade: 6.7/10.0

Projects

Valorant Game Data Pipeline | Individual Case Study | Python, PostgreSQL, dbt, Docker, SQLAlchemy, pytest

- Built to engineer a production-pattern ELT warehouse around Valorant gameplay data, replacing absent match telemetry with a statistically realistic simulation engine covering match timelines, rank progression, and per-round credit economy.
- Designed a modular Python ELT pipeline extracting metadata from the Valorant API, simulating ~10,220 player-match records across 1,000 players and loading 15 raw tables into a Dockerised PostgreSQL warehouse; modelled data through dbt staging views and a governed star-schema mart layer covering player performance, agent pick rates, map win rates, rank distribution, and credit economy.
- Delivered 24 dbt models and 87 passing test nodes validating referential integrity, key uniqueness, and accepted values across all layers, producing a fully governed, self-serve analytics warehouse with zero test failures on build.

Multi-Channel Brand Performance Analytics | Vision Group Retail Inc. | SQL, Dimensional Modeling, Power BI, DAX

- Built to consolidate fragmented brand performance data across multiple sales channels into a single governed analytics layer, eliminating inconsistent metrics and enabling self-serve access for 40+ stakeholders across sales, marketing, and operations.
- Designed star-schema dimensional models and a self-serve semantic layer using Power BI and DAX, applying metric standardisation and documentation practices aligned with analytics engineering governance principles.
- Delivered a trusted, governed KPI layer across 1.2M+ rows that eliminated cross-channel reporting discrepancies and reduced ad hoc reporting requests, enabling consistent metric consumption without data team involvement.

OTT Subscriber Churn Prediction Pipeline | Individual Case Study | Python, AWS (Glue, SageMaker, S3), Flask

- Built to operationalise subscriber churn risk identification for an OTT platform, requiring end-to-end pipeline design from raw event ingestion through feature engineering to production-ready model serving.
- Python (scikit-learn, pandas) for modelling, AWS Glue for ETL transformation, SageMaker for training and deployment, S3 for storage, Flask API for model serving as a consumable data product.
- Achieved approximately 0.75 TPR at 0.20 FPR across scalable analytical datasets, delivering a production-ready churn signal enabling targeted retention interventions for high-risk subscriber segments.

Skills

- **Data Pipeline Engineering:** Apache Kafka | Apache Flink | Apache Spark | PySpark | Apache Airflow | ETL/ELT Pipelines | Batch Ingestion | Near-Real-Time Data Ingestion | Workflow Orchestration | Event-Driven Architecture | Distributed Processing | High Availability Systems
- **Analytics Engineering and Modeling:** dbt | dbt Tests and Source Freshness | Dimensional Modeling | Data Marts | Semantic Layer | Data Contracts | Kimball Methodology | Curated Reporting Layers | Governed Datasets
- **Cloud and Data Platforms:** Snowflake | Databricks | AWS (Glue, Lambda, S3) | GCP (BigQuery) | Azure SQL | PostgreSQL | Docker
- **Data Quality and Governance:** Automated Data Validation | Schema Testing | SLA Monitoring | Pipeline Observability | Audit Controls | RBAC and Access Controls | Data Governance | CDC (Change Data Capture)
- **Languages:** Python | SQL | PySpark | Snowflake SQL | Databricks SQL | DAX
- **Software Engineering:** Clean Code Principles | Modular Architecture | Refactoring | Software Maintainability | Production Systems | Integration Design | API Integration | Data Contracts | Object-Oriented Programming | SDLC
- **Testing:** Unit Testing | Integration Testing | Regression Testing | Test Automation | Automated Validation Frameworks
- **CI/CD and Infrastructure:** GitLab CI/CD | GitHub Actions | Azure DevOps | Terraform
- **Consumption Layer:** Power BI | Tableau | Looker | Sigma
- **Soft Skills:** Cross-Functional Engineering Collaboration | Stakeholder Communication | Independent Project Ownership | Requirements Gathering and Technical Scoping | Code Reviews | Agile Delivery

Certifications

Data Engineer Associate | DataCamp

March 2025 to March 2027

ETL Testing | Udemy

April 2024 to Present

Data Warehousing | Udemy

February 2024 to Present